

Methods

The shark species examined in this study were selected from Ebert, Dando & Fowler, 2014, which offered a comprehensive list of known species globally. We conducted species-specific searches using Google Scholar to identify relevant peer-reviewed studies, agency reports, and graduate student theses published up until 2021. Based on a preliminary examination of the literature, our search terms included all combinations of the scientific or common name of the species, and the following: “social”, “aggregation”, “gathering”, “grouping”, “pack hunting”, “mating”, “breeding”, “shoaling”, “behavior”, “cognition”, “personality”, “circling”. Each source of data was examined for relevant data by at least two independent researchers.

We collected data from all literature sources in which social groups, non-social aggregations, unknown co-occurrences (*i.e.* unclear whether it was a social or non-social), and/or solitary behaviors (hereafter, collectively “behavioral data”) were reported (as defined in McInturf et al. 2023). Given that the term aggregation was most widely used in shark social behavior literature without a standardized definition, we determined these categories based on supporting evidence from descriptions of the co-occurrences within the studies. For example, if the term aggregation was used to indicate preferential association with conspecifics, we classified this paper as evidence for social grouping in that species. Data from juveniles were excluded to control for the possibility for ontogenetic shifts in behavior (see Guttridge et al. 2012 for example), though we highly recommend future studies focus on this topic. We also excluded studies where behaviors of interest could be attributed to human influence only (*i.e.* chumming, aquaculture). In cases of captive research, studies were only included if they reported behaviors that had also been observed in the field. Finally, we excluded data from sources where behaviors were inferred strictly from catch data, as it was impossible to determine the spatial and temporal scales at which individuals were captured.

We also collected data on biological and ecological traits associated with social behavior and/or aggregations in sharks and other taxa. We considered habitat preference, depth occupation, evidence of site fidelity, and mean body length. We used any evidence of site fidelity in each shark species (*i.e.*, year round residency, seasonal site fidelity including breeding site fidelity) and categorized site fidelity as a binary variable (*i.e.* yes or no). We also gathered biological and ecological data from FishBase (Froese & Pauly, 2019), using standardized descriptions for habitat preference (bathydemersal, benthopelagic, demersal, pelagic, bathypelagic, or reef-associated), depth occupation (epipelagic, mesopelagic, and bathypelagic), and mean body length. Habitat preference data was mutually exclusive while species were placed in multiple categories for depth.